

Poisson distribution

The Poisson distribution is often used in situations where we are counting the number of independent successes in a particular region or interval of time. For example, the number of emails you receive in an hour, or the number of earthquakes in a year in some region of the world.

Definition (Poisson Distribution $X \sim \text{Pois}(\lambda)$)

An r.v. X has the Poisson distribution with parameter λ , where $\lambda > 0$, if the PMF of X is

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!} \quad \text{where } k = 0, 1, 2, \dots$$

$$E[X] = \lambda$$

$$\text{Var}(X) = \lambda$$

X := number of occurrence of an event in a specific interval of time.

λ := is the average rate of occurrence of the events.

Eg 2. Between 1950 and 2020, 140 major earthquakes were recorded in a particular country. Statistical evidence suggests a Poisson distribution. Find the probabilities that:

- (a) 2
- (b) fewer than 2,
- (c) more than 2 major earthquakes will occur in this country in a particular year.
- (d) Also, identify the variance of this probability distribution.

In $2020 - 1950 = 70$ years, 140 major earthquakes occur. Hence rate of occurrence per year $\lambda = \frac{140}{70} = 2$.

$$P(X=k) = e^{-2} \frac{2^k}{k!} \quad k=0,1,2,\dots$$

$$(a) \quad P(X=2) = e^{-2} \frac{2^2}{2!} = 2e^{-2}$$

$$(b) \quad P(X < 2) = P(X=0) + P(X=1) \\ = e^{-2} \frac{2^0}{0!} + e^{-2} \frac{2^1}{1!} = e^{-2}(1+2) = 3e^{-2}$$

$$(c) \quad P(X > 2) = 1 - P(X \leq 2) = 1 - [3e^{-2} + 2e^{-2}] \\ = 1 - 5e^{-2}$$

$$(d) \quad \text{var}(X) = \lambda = 2.$$

Eg 3-A. Traffic accidents at a particular intersection follow Poisson distribution with an average rate of 1.4 per week.

(a) What is the probability that the next week is accident-free?

$$\lambda = 1.4 \quad p(x=0) = e^{-1.4} \frac{(1.4)^0}{0!} = e^{-1.4}$$

(b) What is the probability that there will be exactly 3 accidents next week?

$$\lambda = 1.4 \quad p(x=3) = e^{-1.4} \frac{(1.4)^3}{3!}$$

Eg 3-B. Traffic accidents at a particular intersection follow Poisson distribution with an average rate of 1.4 per week.

(c) What is the probability that there will be at least 2 accidents during the next two weeks?

$$\text{for two weeks } \lambda = 2 \cdot (1.4) = 2.8$$

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = [1 - P(X=0) + P(X=1)] \\ &= 1 - \left[e^{-2.8} \frac{(2.8)^0}{0!} + e^{-2.8} \frac{2.8}{1!} \right] = 1 - 3.8 e^{-2.8} \end{aligned}$$

(d) What is the probability that there will be exactly 2 accidents tomorrow?

$$\lambda = \frac{1.4}{7} = 0.2 \text{ per day}$$

$$P(X=2) = e^{-0.2} \frac{(0.2)^2}{2!} = e^{-0.2} \frac{(0.04)}{2} = \frac{(0.02)}{e^{0.2}}$$

Eg 3-C. Traffic accidents at a particular intersection follow Poisson distribution with an average rate of 1.4 per week.

(e) What is the probability that the next accident will not occur for three days?

$$3 \text{ days} \Rightarrow \lambda = \frac{1.4}{7} \cdot 3 = (0.2) \cdot 3 = 0.6$$

$$p(X=0) = \frac{e^{-0.6} (0.6)^0}{0!} = e^{-0.6}$$

(f) What is the probability that there will be exactly three accident-free weeks during the next eight weeks?

Success = An accident free week

$$p = p(\text{success}) = p(X=0) = \frac{e^{-1.4} (1.4)^0}{0!} = e^{-1.4}$$

Y = number of accident free week during next 8 week

$$Y \sim \text{Bin}(n=8, p = e^{-1.4})$$

$$P(Y=3) = \binom{8}{3} p^3 (1-p)^5$$

$$= \binom{8}{3} (e^{-1.4})^3 (1 - e^{-1.4})^5$$

The following is only for your information, it will not be a part of your homework or grading:

Note: The Poisson distribution is often used in situation where we are counting the number of successes in a particular region or interval of time, and there are a large number of trials, each with a small probability of success.

In otherwords, if $n > 50$ and $np < 5$, then $\text{Bin}(n, p) \approx \text{Pois}(\lambda = np)$.

In modelling, if even the events are weakly dependent or the the probability of the events are not the same, we can also use Poisson for approximation.

For example, assume we have a family of weakly dependent events

$E_1, E_2, \dots, E_n$, each with probability $P(E_i) = p_i$. Let $X = \sum_{i=1}^n I(E_i)$.

When n is very large and p_i s are small. Then $X \approx \text{Pois}(\lambda = \sum_{i=1}^n p_i)$